

XIAMEN UNIVERSITY MALAYSIA

厦門大學馬來西亞分校

COM105 Engineering Physics (I)

Lecture 9: Dynamics of Rotational Motion

By:
Dr. Ng Zi Neng
Email: zineng.ng@xmu.edu.my

In this chapter, you'll learn…

- what is meant by the torque produced by a force.
- how the net torque on an object affects the object's rotational motion.
- how to analyze the motion of an object that both rotates and moves as a whole through space.
- how to solve problems that involve work and power for rotating objects.
- how the angular momentum of an object can remain constant even if the object changes shape.

Torque 扭转力



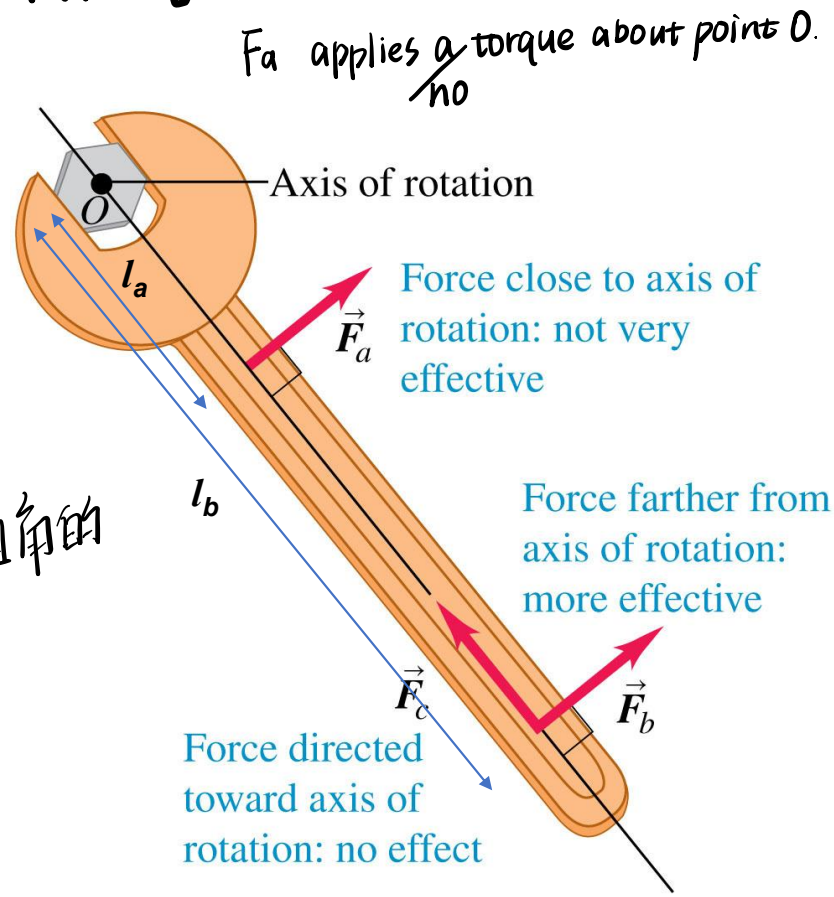
- These jugglers toss the pins so that they rotate in midair.
- What does it take to start a stationary object rotating or to bring a spinning object to a halt?
- We'll introduce some new concepts, such as torque and angular momentum, to deepen our understanding of rotational motion.

力矩是一个物理量，表示的是一个力对所转物体产生的效应

/tɔːk/ Torque 转力矩 / 扭矩

- We say that F_a applies a torque about point O to the wrench, F_b applies a **greater** torque about O, and F_c applies zero torque about O.
- That is because torque also depends on the **perpendicular** distance (l_a or l_b) between point O and the line of action of the force, known as the lever or moment arm of force (F_a or F_b), hence

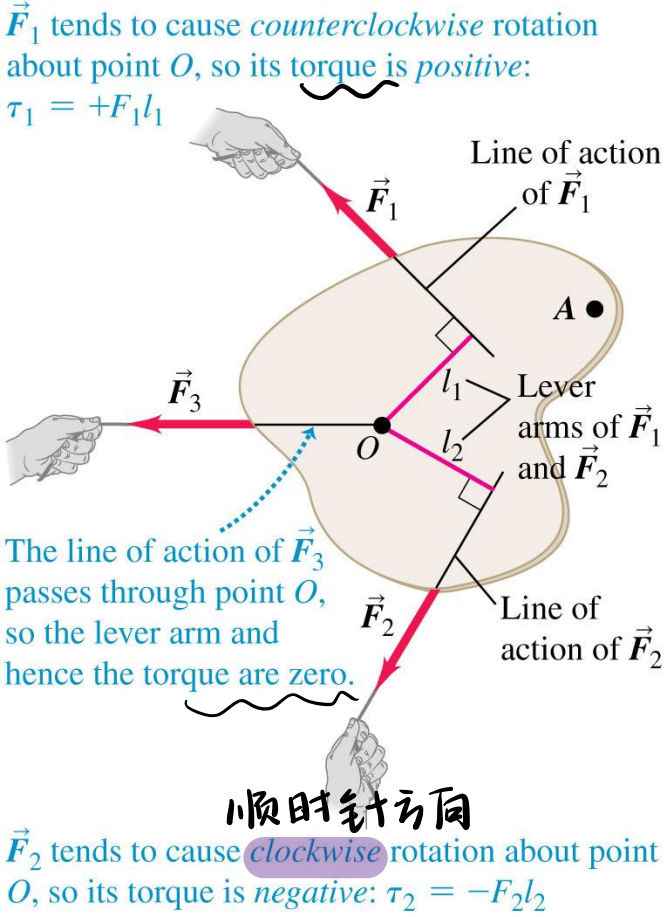
$$\tau = Fl \qquad (10.1)$$



clockwise
counterclockwise

Torque

逆时针方向，从右向左



- To distinguish between the counterclockwise rotation caused by F_1 and clockwise rotation caused by F_2, we need to choose a positive sense of rotation.
- With the choice that counterclockwise torques are positive and clockwise torques are negative, the torques of F_1 and F_2 about O are

$$\tau_1 = +F_1 l_1 \qquad \tau_2 = -F_2 l_2$$

- The SI unit of torque is the newton-meter. In our discussion of work and energy we called this combination the joule. But torque is **not** work or energy, and torque should be expressed in newton-meters, not joules.

SI unit : N·m.

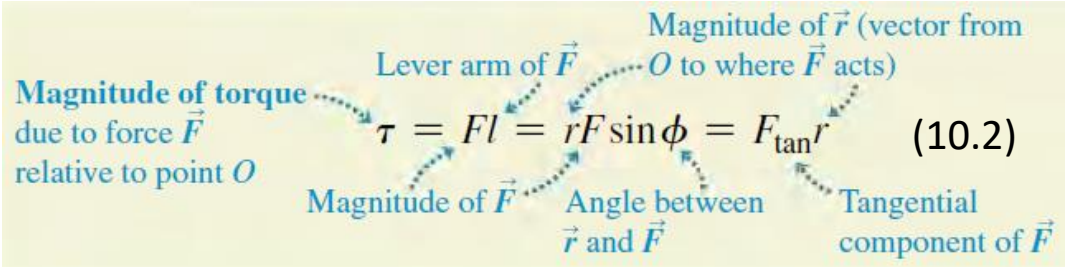
lever arm

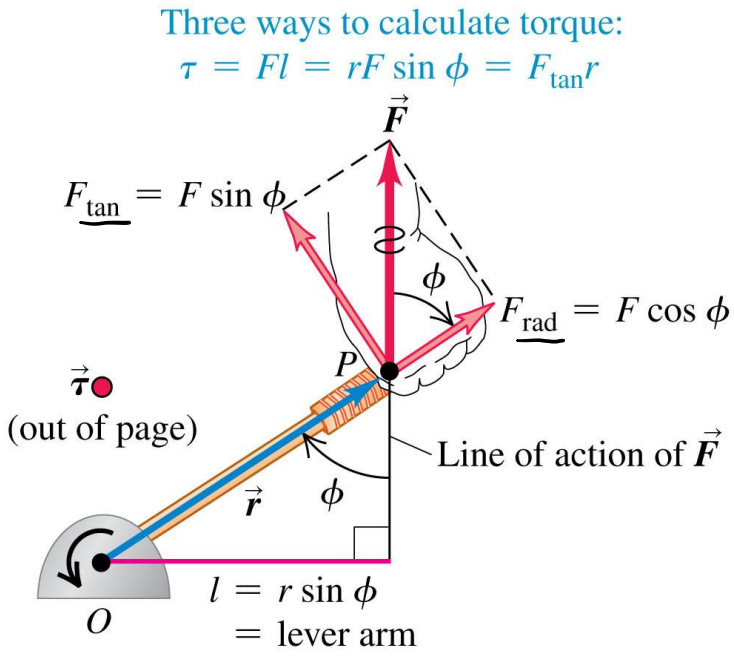
Torque

To calculate the torque of ***F***:

- Find the lever arm ***l*** and use $\tau = Fl$.
- Determine the angle between the vectors ***r*** and ***F***; the lever arm is $r \sin\phi$, so $\tau = r F \sin\phi$.
- ***F*** can also be represented in terms of radial component F_{rad} along the direction of ***r*** and a tangential component F_{tan} at right angles, perpendicular to ***r***. Then

$$F_{tan} = F \sin\phi$$





$\tau = F_{tan} r$ 可以分解F/l
$= rF\sin\phi$

Torque

- Torque is the vector product of **r** and **F**:

叉乘，结果大小是所形成四边形S

Torque vector due to force $\vec{F}$ relative to point O ... $\vec{\tau} = \vec{r} \times \vec{F}$ ... Force $\vec{F}$

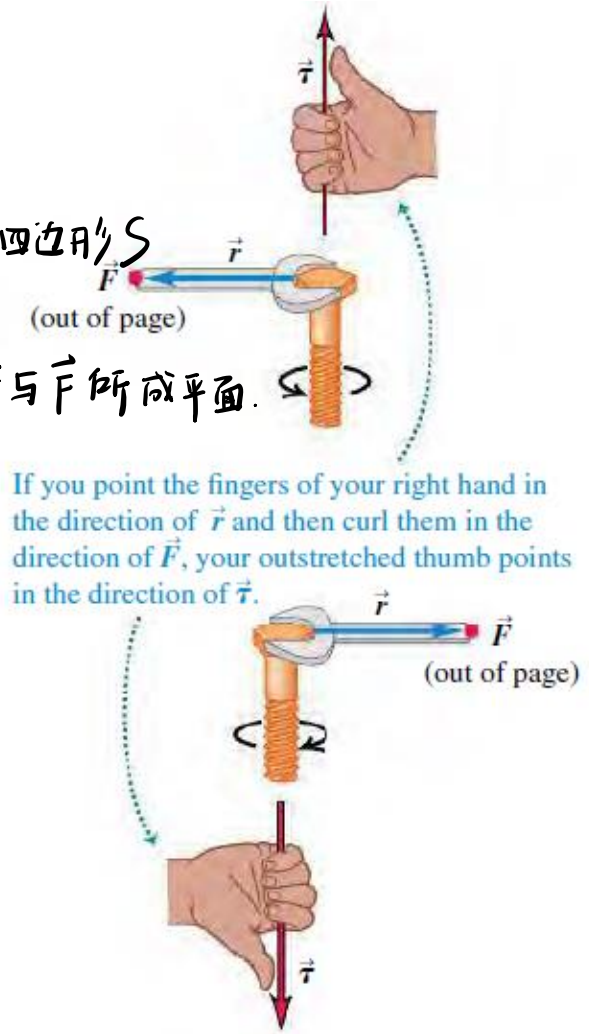
Vector from O to where $\vec{F}$ acts

结果方向是垂直于$\vec{r}$与$\vec{F}$所成平面.

用右手.

- Use right hand rule to determine the direction of torque:

右手定则来定义τ的方向

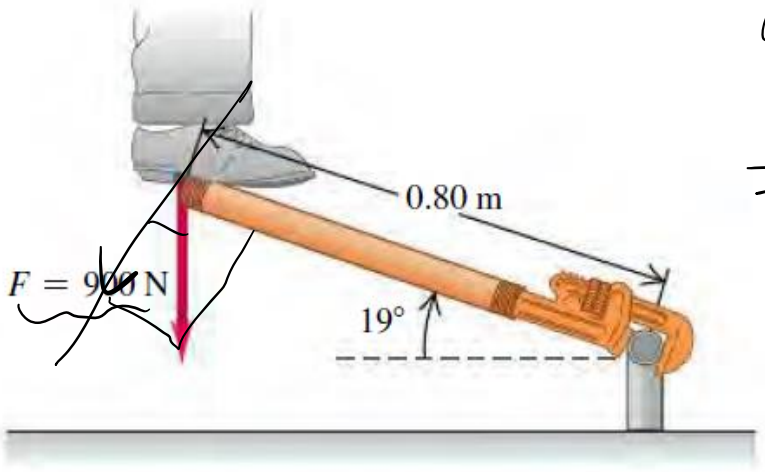


Torque

EXAMPLE 10.1 APPLYING A TORQUE

To loosen a pipe fitting, a plumber slips a piece of scrap pipe (a "cheater") over his wrench handle. He stands on the end of the cheater, applying his 900-N weight at a point 0.80 m from the center of the fitting (**Fig. 10.5a**). The wrench handle and cheater make an angle of 19° with the horizontal. Find the magnitude and direction of the torque he applies about the center of the fitting.

(a) Diagram of situation

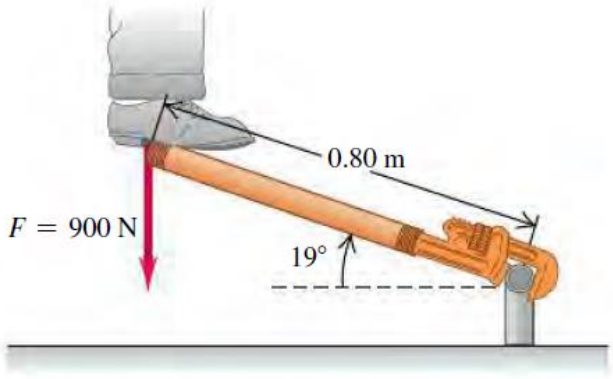


$900 \times \cos 19^\circ \times 0.8$

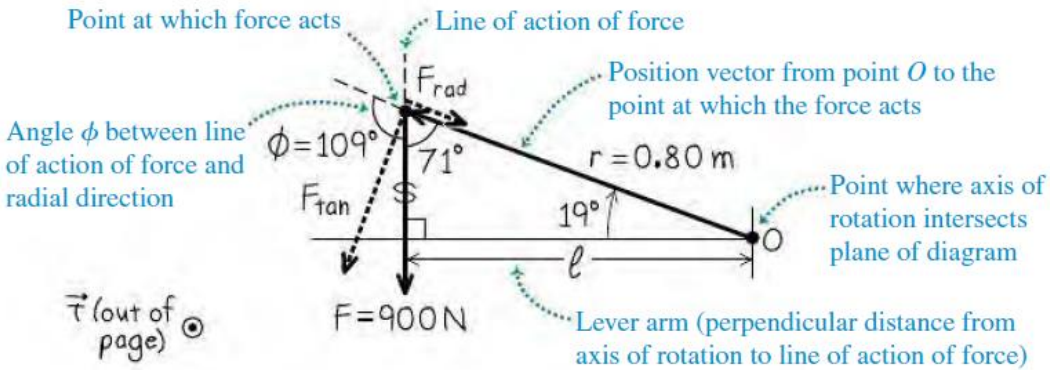
$= 680.77$

Torque

(a) Diagram of situation



(b) Free-body diagram



SOLUTION

IDENTIFY and SET UP: Figure 10.5b shows the vectors $\vec{r}$ and $\vec{F}$ and the angle between them ($\phi = 109°$). Equation (10.1) or (10.2) will tell us the magnitude of the torque. The right-hand rule with Eq. (10.3), $\vec{\tau} = \vec{r} \times \vec{F}$, will tell us the direction of the torque.

EXECUTE: To use Eq. (10.1), we first calculate the lever arm l. As Fig. 10.5b shows,

$$l = r\sin\phi = (0.80\text{ m})\sin 109° = 0.76\text{ m}$$

Then Eq. (10.1) tells us that the magnitude of the torque is

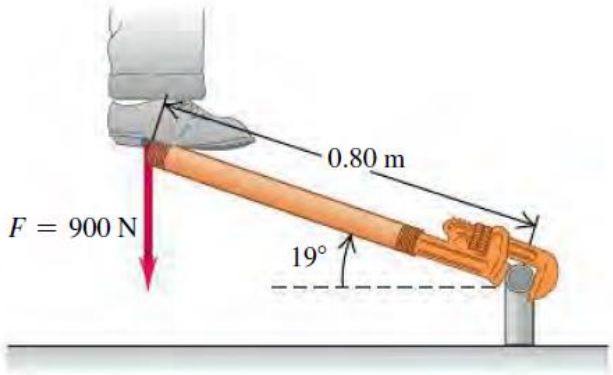
$$\tau = Fl = (900\text{ N})(0.76\text{ m}) = 680\text{ N}\cdot\text{m}$$

We get the same result from Eq. (10.2):

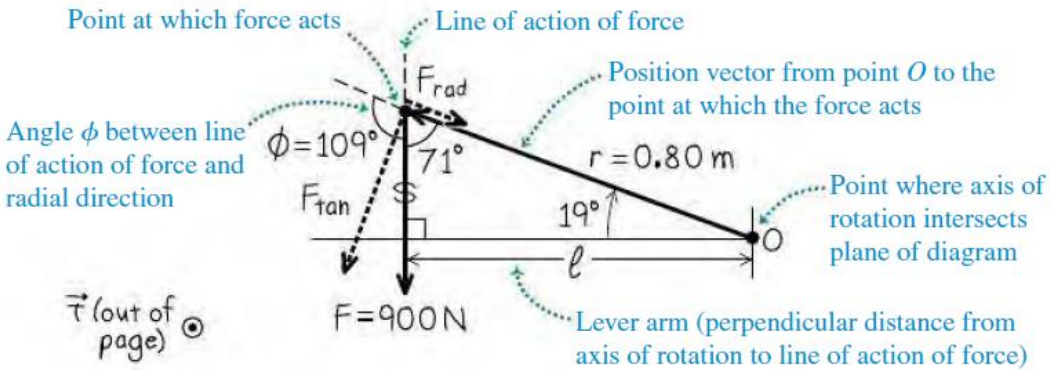
$$\tau = rF\sin\phi = (0.80\text{ m})(900\text{ N})(\sin 109°) = 680\text{ N}\cdot\text{m}$$

Torque

(a) Diagram of situation



(b) Free-body diagram



Alternatively, we can find $F_{\tan}$, the tangential component of $\vec{F}$ that acts perpendicular to $\vec{r}$. Figure 10.5b shows that this component is at an angle of $109° - 90° = 19°$ from $\vec{F}$, so $F_{\tan} = F(\cos 19°) = (900\ \text{N})(\cos 19°) = 851\ \text{N}$. Then, from Eq. 10.2,

$$\tau = F_{\tan} r = (851\ \text{N})(0.80\ \text{m}) = 680\ \text{N} \cdot \text{m}$$

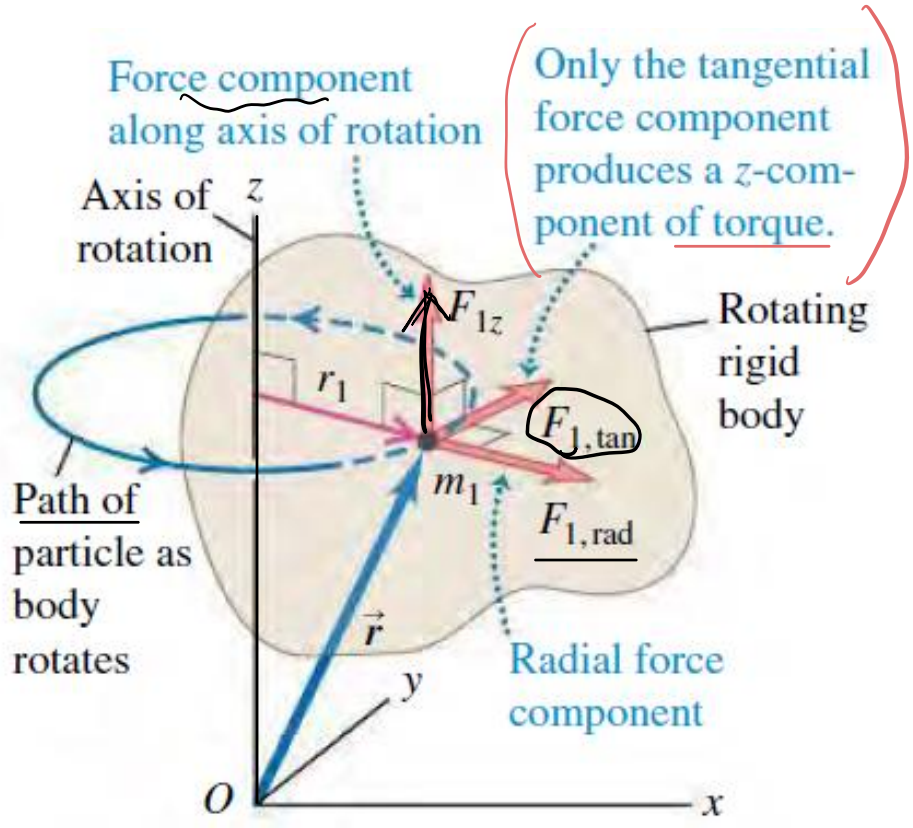
Curl the fingers of your right hand from the direction of $\vec{r}$ (in the plane of Fig. 10.5b, to the left and up) into the direction of $\vec{F}$ (straight down). Then your right thumb points out of the plane of the figure: This is the direction of $\vec{\tau}$.

EVALUATE: To check the direction of $\vec{\tau}$, note that the force in Fig. 10.5 tends to produce a counterclockwise rotation about O. If you curl the fingers of your right hand in a counterclockwise direction, the thumb points out of the plane of Fig. 10.5, which is indeed the direction of the torque.

Torque and Angular Acceleration

- As a rigid body rotates around the z-axis, a net force F_1 acts on one particle of the body.
- Only the force component $F_{1,tan}$ can affect the rotation, because only $F_{1,tan}$ exerts a torque about O with along the rotation axis.

有且仅有 tangent 方向作用扭矩



Torque and Angular Acceleration

- Newton's second law for the tangential component is:

$$F_{1,\tan} = m_1 a_{1,\tan} \quad (10.4)$$

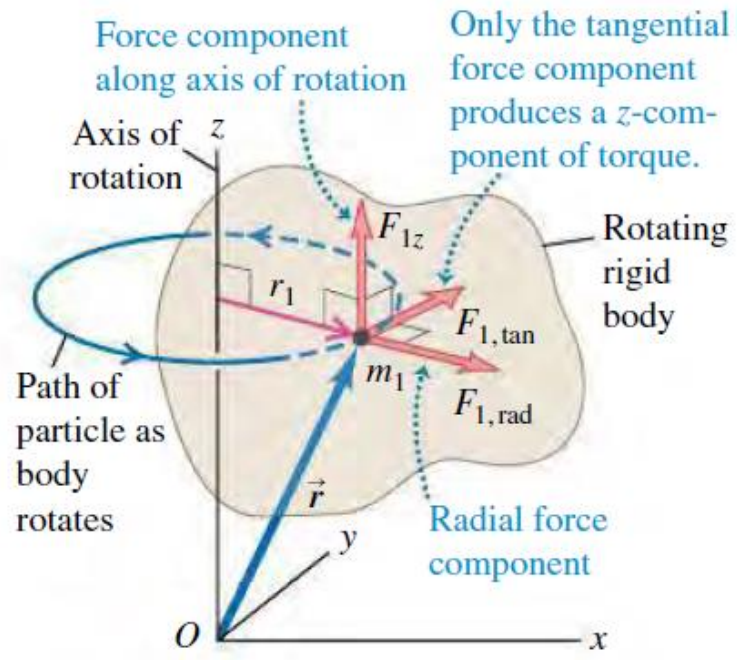
- Since $a_{1,\tan} = r_1 \alpha_z$ (9.14) and $\tau_{1z} = F_{1,\tan} r_1$ (10.2), then

$$\tau_{1z} = m_1 r_1^2 \alpha_z \quad (10.5)$$

- We write such an equation for every particle in the body, then add all these equations:

$$\Sigma \tau_{iz} = \left(\Sigma m_i r_i^2 \right) \alpha_z \quad (10.6)$$

where inertia $I = \Sigma m_i r_i^2$



Torque and Angular Acceleration

- Giving us the final equation, which is the rotational analog of Newton's second law for a rigid body:

Rotational analog of Newton's second law for a rigid body:

$$\sum \tau_z = I\alpha_z \qquad (10.7)$$

Net torque on a rigid body about z-axis

Moment of inertia of rigid body about z-axis

Angular acceleration of rigid body about z-axis

- Loosening or tightening a screw requires giving it an angular acceleration and hence applying a torque.

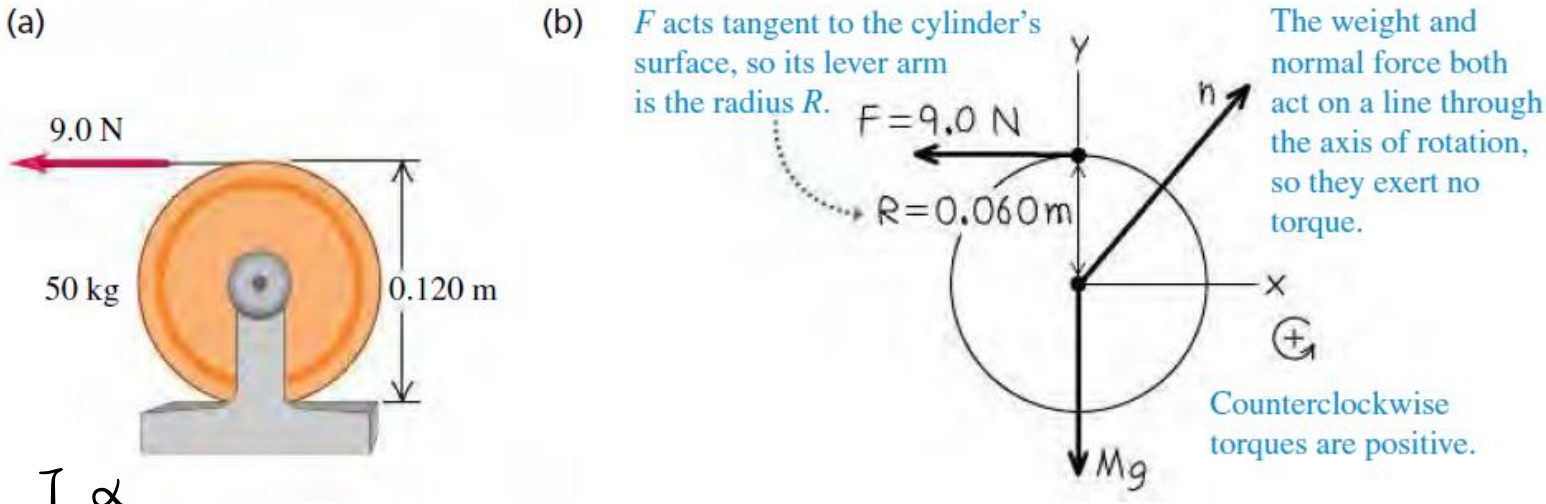


Torque and Angular Acceleration

EXAMPLE 10.2 AN UNWINDING CABLE I

What is the cable's acceleration?

10.9 (a) Cylinder and cable. (b) Our free-body diagram for the cylinder.



$9 \times R = \tau = I\alpha$

0.06

$0.54 = \quad (0.06)^2 \cdot \alpha$

$\alpha = 6$

Torque and Angular Acceleration

$a = \alpha r = 0.36\ m/s^2$

SOLUTION

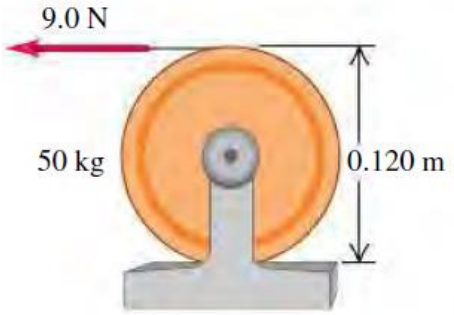
EXECUTE: The lever arm of F is equal to the radius $R = 0.060$ m of the cylinder, so the torque is $\tau_z = FR$. (This torque is positive, as it tends to cause a counterclockwise rotation.) From Table 9.2, case (f), the moment of inertia of the cylinder about the rotation axis is $I = \frac{1}{2}MR^2$. Then Eq. (10.7) tells us that

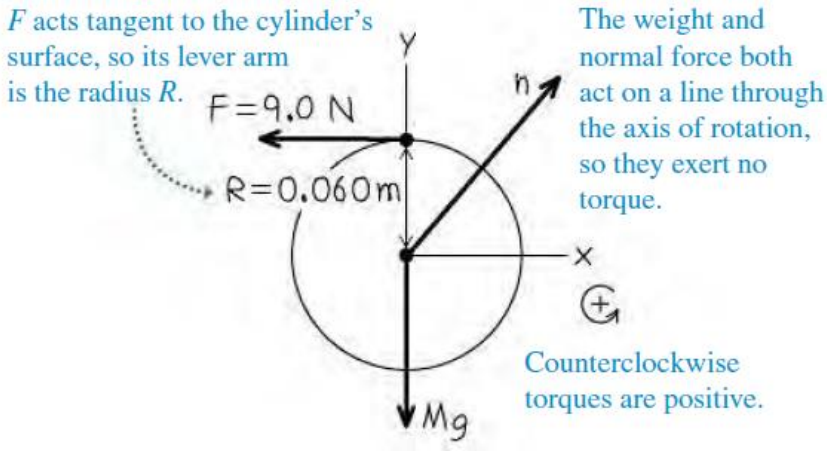
$$\alpha_z = \frac{\tau_z}{I} = \frac{FR}{MR^2/2} = \frac{2F}{MR} = \frac{2(9.0\text{ N})}{(50\text{ kg})(0.060\text{ m})} = 6.0\text{ rad/s}^2$$

(We can add "rad" to our result because radians are dimensionless.)

To get the linear acceleration of the cable, recall from Section 9.3 that the acceleration of a cable unwinding from a cylinder is the same as the tangential acceleration of a point on the surface of the cylinder where the cable is tangent to it. This tangential acceleration is given by Eq. (9.14):

$$a_{\text{tan}} = R\alpha_z = (0.060\text{ m})(6.0\text{ rad/s}^2) = 0.36\text{ m/s}^2$$





有一个移动的Axis，Rigid-Body Rotation.

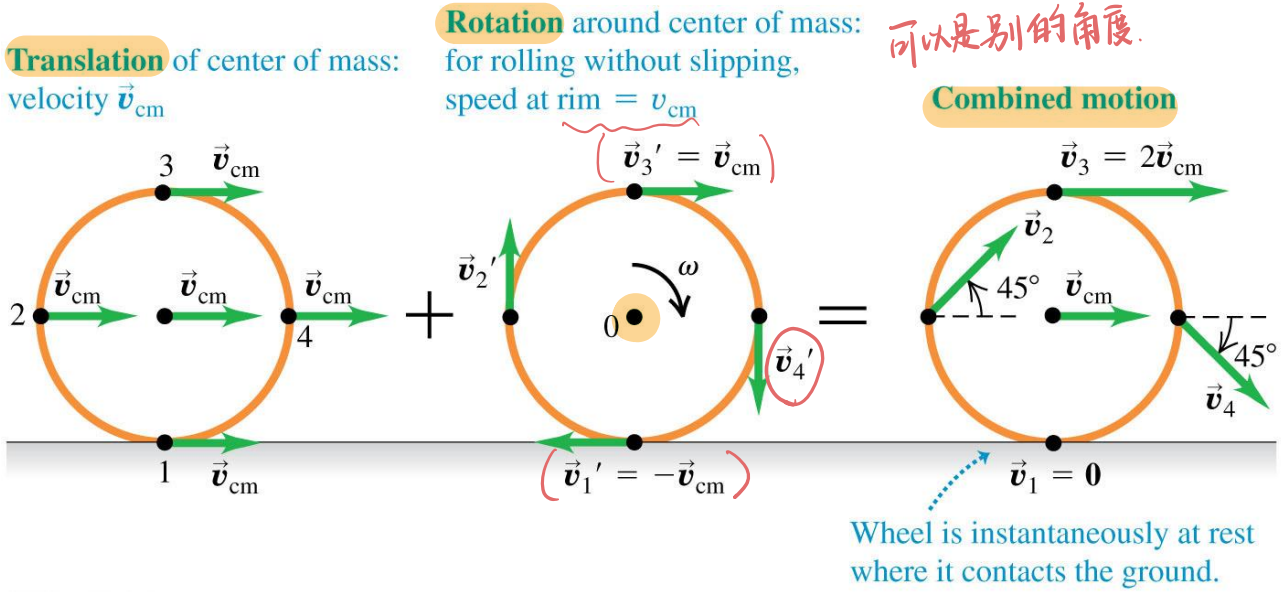
Rigid-Body Rotation About a Moving Axis

- The motion of a rolling wheel is the combination of the translational motion of the center of mass and the rotational motion of the wheel around the center of mass.

有 v_{cm} 和 $v_{angular}$ 两个合成在一起

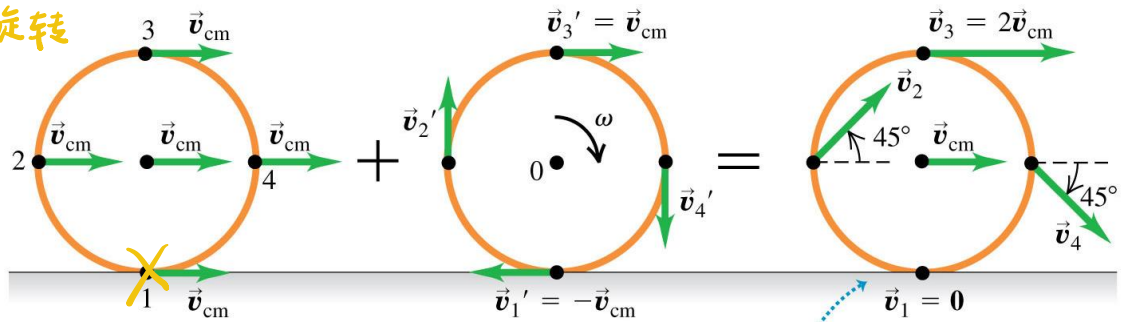
- Translational motion is the motion in which all points of a moving body move uniformly in the same line or direction (the motion of the body as a whole through space)

看做质点的整体速度
+
扭转的速度
=
整体的速度.



Rigid-Body Rotation About a Moving Axis

相当于围绕着点1的旋转



- At any instant we can think of the wheel as rotating about an "instantaneous axis" of rotation that passes through the point of contact with the ground.
- The angular velocity ω is the same for this axis as for an axis through the center of mass.
- The kinetic energy of the wheel is $K = \frac{1}{2} I_1 \omega^2$, where I_1 is the moment of inertia of the wheel about an axis through point 1.
- In the parallel-axis theorem, Eq. (9.19), $I_1 = I_{cm} + MR^2$, where M is the total mass of the wheel and I_{cm} is the moment of inertia with respect to an axis through the center of mass.

$$K = \tfrac{1}{2} I_1 \omega^2 = \tfrac{1}{2} I_{cm} \omega^2 + \tfrac{1}{2} M R^2 \omega^2 = \tfrac{1}{2} I_{cm} \omega^2 + \tfrac{1}{2} M v_{cm}^2$$

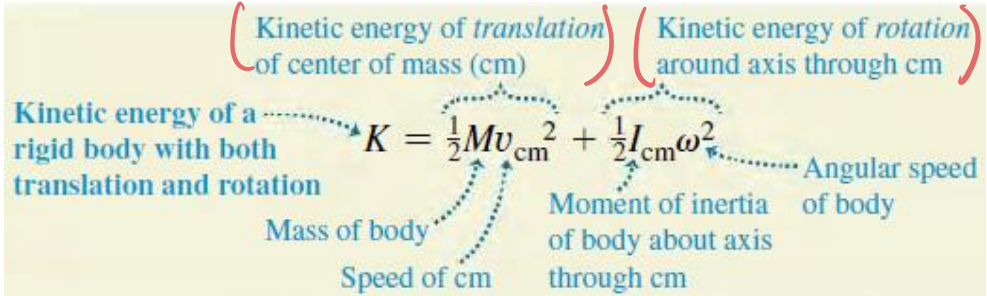
$K = \frac{1}{2}mv^2$ $= \frac{1}{2}m \cdot \omega^2 r^2$ $= \frac{1}{2} I r^2$

$v = \omega r$

Rigid-Body Rotation About a Moving Axis

- Hence, the kinetic energy K of a rigid body that has both translational and rotational motions. For such a body, K is the sum of two parts:

cm : center mass

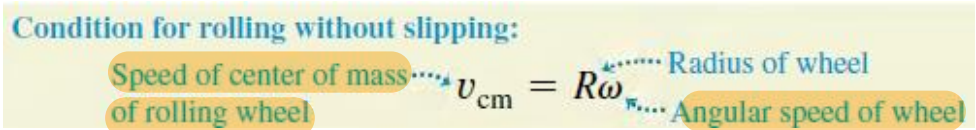


关于速度，是Vtan 与 Vcm 的矢量合成

(10.8)

关于K，是 Kcm 与 Krotate 的标量相加

- The condition for rolling without slipping is



(10.11) Vcm = Rω 两个系统的 bridge.

- The gravitational potential energy has to be considered if a rigid body changes height as it moves,

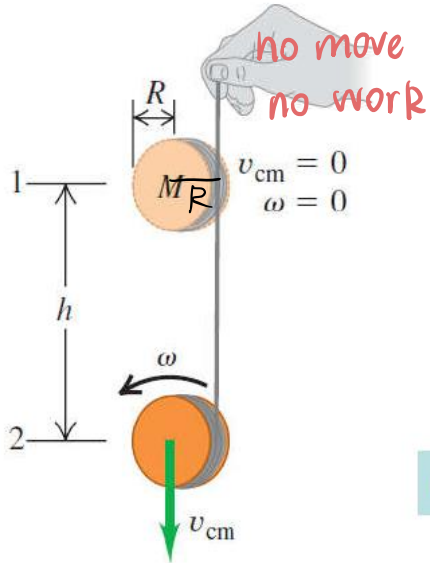
$$U = Mgy_{cm}$$

- Total Energy: $K + U$

Rigid-Body Rotation About a Moving Axis

Example 10.4 Speed of a primitive yo-yo

You make a primitive yo-yo by wrapping a massless string around a solid cylinder with mass M and radius R (Fig. 10.15). You hold the free end of the string stationary and release the cylinder from rest. The string unwinds but does not slip or stretch as the cylinder descends and rotates. Using energy considerations, find the speed v_{cm} of the center of mass of the cylinder after it has descended a distance h.



The upper end of the string is held fixed, not pulled upward, so your hand does no work on the string–cylinder system. There is friction between the string and the cylinder, but the string doesn't slip so no mechanical energy is lost.

⇩

Conservation of mechanical energy

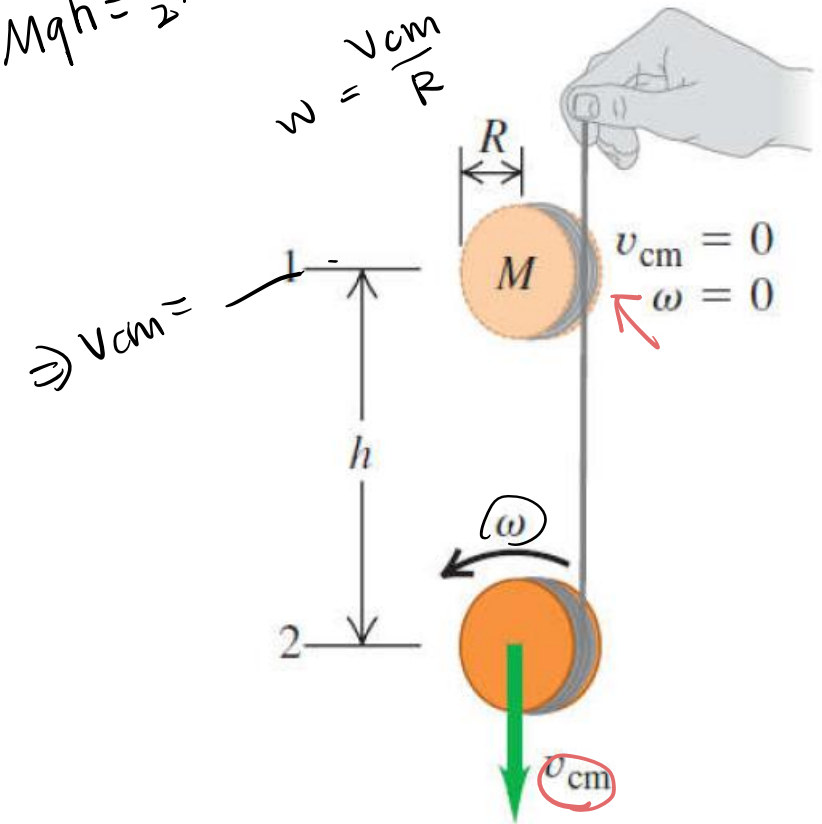
关于此题在能量角度上的解法

Rigid-Body Rotation About a Moving Axis

Solution:

$K_1 = 0$, $U_1 = Mgh$

$I = \frac{1}{2}MR^2$ $\omega = v_{cm}/R$



$$K_2 = \tfrac{1}{2}Mv_{cm}^2 + \tfrac{1}{2}\left(\tfrac{1}{2}MR^2\right)\left(\frac{v_{cm}}{R}\right)^2$$

$$= \tfrac{3}{4}Mv_{cm}^2$$

Using conservation of mechanical energy,

$$K_1 + U_1 = K_2 + U_2$$

$$0 + Mgh = \tfrac{3}{4}Mv_{cm}^2 + 0$$

$$v_{cm} = \sqrt{\tfrac{4}{3}gh}$$

Combined Translation and Rotation: Dynamics

动力学

- For an extended body, the acceleration of the center of mass is the same as that of a particle of the same mass acted on by all the external forces on the actual body:

拉伸实体 a_{cm}

对于整体

Net external force on a body: $\sum \vec{F}_{ext} = M\vec{a}_{cm}$ (10.12)

(M: Total mass of body; $\vec{a}_{cm}$: Acceleration of center of mass)

Net external force on a body, 等效在一个点上的整体合力. $= M\,\vec{a}_{cm}$.

- The rotational motion about the center of mass is described by the rotational analog of Newton's second law,

模拟

对于扭转力. 扭矩

Net torque on a rigid body about z-axis through center of mass: $\sum \tau_z = I_{cm}\alpha_z$ (10.13)

(I_{cm}: Moment of inertia of rigid body about z-axis; α_z: Angular acceleration of rigid body about z-axis)

- With two conditions:
  1. The axis through the center of mass must be an axis of symmetry. /ˈsɪmətri/
  2. The axis must not change direction.

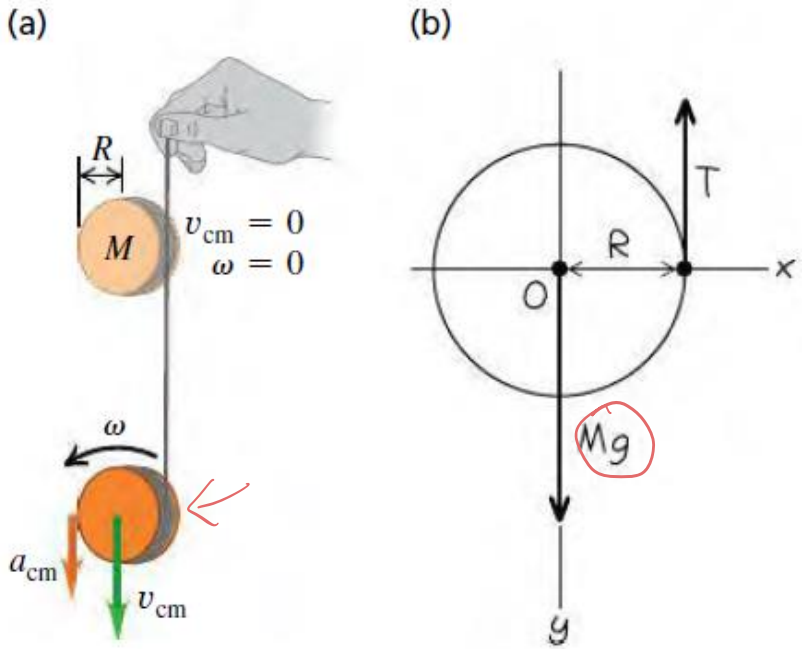
n 对称的

轴是物体中心
不能在沿轴方向上有运动

Combined Translation and Rotation: Dynamics

EXAMPLE 10.6 ACCELERATION OF A PRIMITIVE YO-YO

For the primitive yo-yo in Example 10.4 (**Fig. 10.18a**), find the downward acceleration of the cylinder and the tension in the string.



Combined Translation and Rotation: Dynamics

关于此题在力的角度上的解法

SOLUTION

EXECUTE: From Eq. (10.12),

对整体，是合在一点的矢量和

$$\Sigma F_y = Mg + (-T) = Ma_{\text{cm-}y} \qquad (10.14)$$

From Eq. (10.13),

会给或 对于扭矩，是 切向力 × 力矩

$$\Sigma \tau_z = TR = I_{\text{cm}}\alpha_z = \tfrac{1}{2}MR^2\alpha_z \qquad (10.15)$$

也是 $I_{cm}\alpha_z$ (仿牛二)

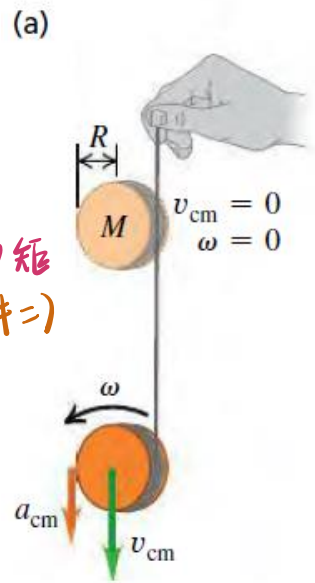
From Eq. (10.11), $v_{\text{cm-}z} = R\omega_z$; the derivative of this expression with respect to time gives us

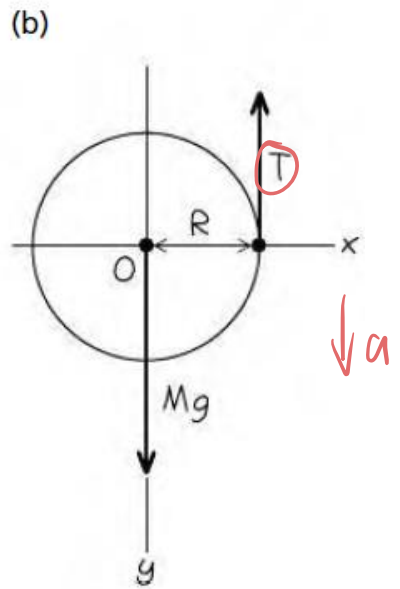
$$a_{\text{cm-}y} = R\alpha_z \qquad (10.16)$$

We now use Eq. (10.16) to eliminate α_z from Eq. (10.15) and then solve Eqs. (10.14) and (10.15) simultaneously for T and $a_{\text{cm-}y}$:

$$a_{\text{cm-}y} = \tfrac{2}{3}g \qquad T = \tfrac{1}{3}Mg$$

EVALUATE: The string slows the fall of the yo-yo, but not enough to stop it completely. Hence $a_{\text{cm-}y}$ is less than the free-fall value g and T is less than the yo-yo weight Mg.





$\alpha_z = a_{cm}/R$

$T \cdot R = \frac{1}{2}MR^2\left(\frac{a_{cm}}{R}\right)$

$T = \frac{1}{2}Ma_{cm}$

$Mg - \frac{1}{2}Ma_{cm} = Ma$

$a = \frac{2}{3}g$

Combined Translation and Rotation: Dynamics

Example 10.7 Acceleration of a rolling sphere

滚下去的，不是滑下去　斜坡

A bowling ball rolls without slipping down a ramp, which is inclined at an angle β to the horizontal (Fig. 10.19a). What are the ball's acceleration and the magnitude of the friction force on the ball? Treat the ball as a uniform solid sphere, ignoring the finger holes.

倾斜　a　f　习阻球体

$F_N = Mg\cos\theta$

$\sum F = Mg\sin\theta - f = M a_{cm-x}$

$Mg\sin\beta - f = Ma$

$f = Mg\sin\beta - Ma$

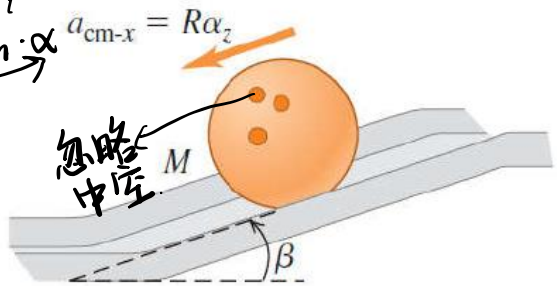
$f = I_{cm} \cdot \frac{a_{cm}}{R^2}$

$\sum \tau = f_s \cdot R - Mg\sin\beta \cdot 0 = f_s \cdot R = I_{cm} \cdot \alpha$

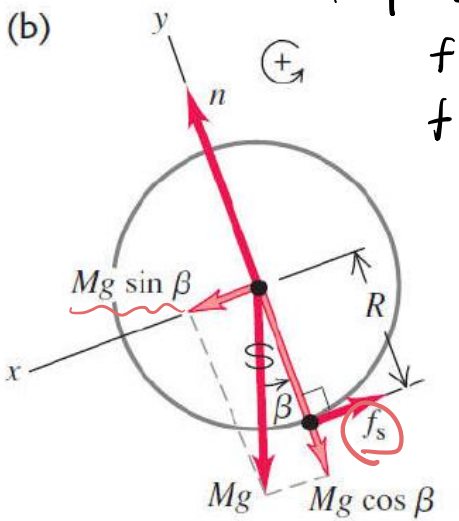
$a_{cm} = R \cdot \alpha_z$

$\alpha = a_{cm}/R$

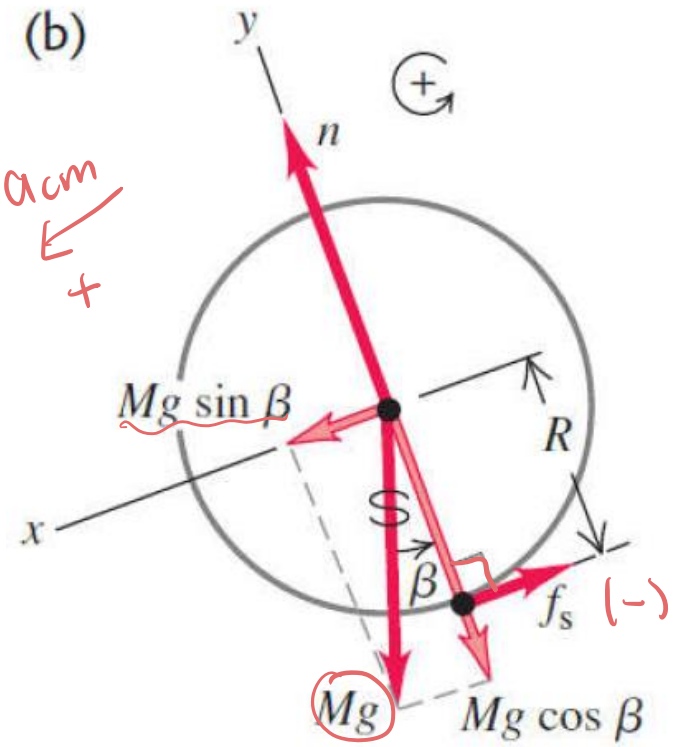
(a)



(b)



Combined Translation and Rotation: Dynamics



The ball's moment of inertia is $I_{cm} = \frac{2}{5}MR^2$.

$$\sum F_x = Mg \sin \beta + (-f) = Ma_{cm\text{-}x} \quad (10.17)$$

$$\sum \tau_z = fR = I_{cm}\alpha_z = \left(\tfrac{2}{5}MR^2\right)\alpha_z \quad (10.18)$$

The ball rolls without slipping,

$$a_{cm\text{-}x} = R\alpha_z$$

$$fR = \tfrac{2}{5}MRa_{cm\text{-}x}$$

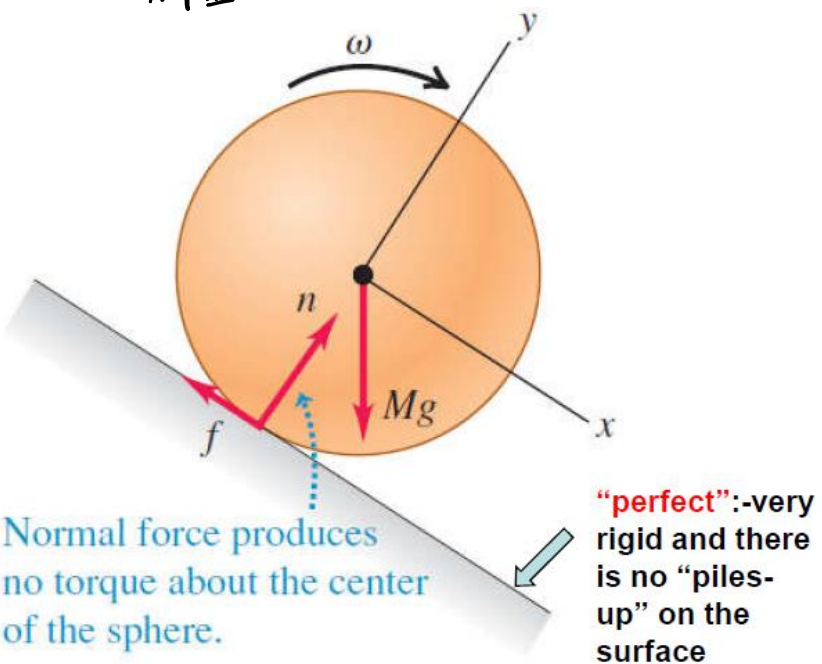
$$a_{cm\text{-}x} = \tfrac{5}{7}g \sin \beta$$

Roll fiction 有两种情况.

Rolling friction 滚动摩擦

- Case 1: perfectly rigid sphere rolling on a perfectly rigid surface (incline/slope)

斜面

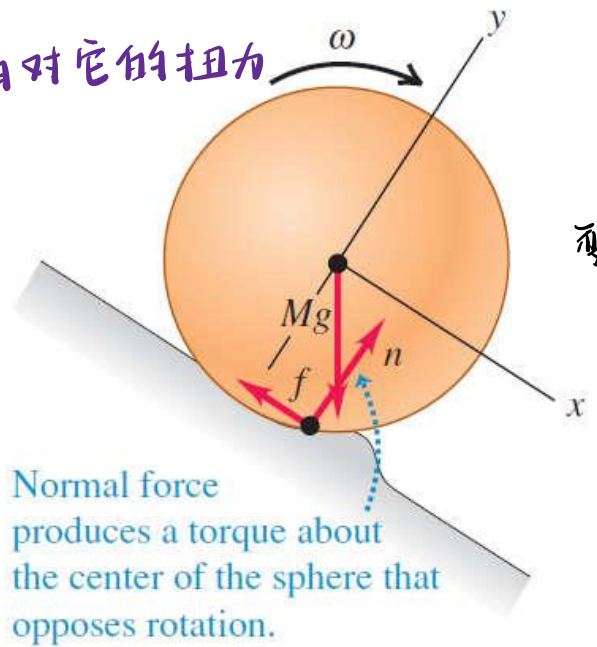


- Assume that the rolling body and the surface over which it rolls are perfectly rigid.
- The line of action of the normal force passes through the center of the sphere, so its torque is zero.
- Hence, there is no sliding 滑动 at the point of contact, so **the friction force does no work.**

Rolling friction

- Case 2: perfectly rigid sphere rolling on a deformed surface (incline/slope)

此时支持有对它的扭力



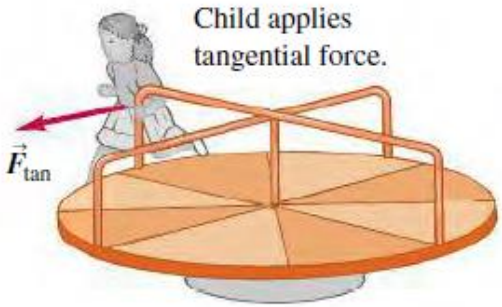
- The normal force now exerts a torque that opposes the rotation.
- In addition, there is some sliding of the sphere over the surface due to the deformation, causing mechanical energy to be lost. 有碰撞，有能量损耗
- The combination of these two effects is the phenomenon of rolling friction.
- Rolling friction also occurs if the rolling body is deformable, such as an automobile tire. 可变形的

- **Generally, the rolling body and the surface are rigid enough that it is regarded as a rolling motion without slipping, so the work done by the friction is zero (Case A).**

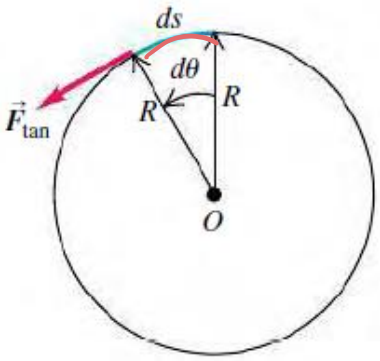
Work and Power in Rotational Motion

10.21 A tangential force applied to a rotating body does work.

(a)



(b) Overhead view of merry-go-round



- A child running while pushing on a playground merry-go-round, a tangential force $\boldsymbol{F}_{tan}$ acts at the rim of a pivoted disk.
- The disk rotates through an extremely small angle $d\theta$ about a fixed axis during an extremely small time interval dt.
- The work dW done by the force $\boldsymbol{F}_{tan}$ while a point on the rim moves a distance ds is $dW = F_{tan}\, ds$. Since $d\theta$ is measured in radians, $ds = R\, d\theta$, then

$$dW = F_{\tan} R\, d\theta$$

- Now $F_{tan}\, R$ is the torque τ_z due to the force $\boldsymbol{F}_{tan}$, so

$$dW = \tau_z\, d\theta \qquad (10.19)$$

- As the disk rotates from θ_1 to θ_2, the total work done by a constant torque is

Work done by a constant torque τ_z: $W = \tau_z(\theta_2 - \theta_1) = \tau_z \Delta\theta$ (10.21)

(Torque; Final minus initial angular position = angular displacement)

$\Rightarrow W = \tau_z (\theta_2 - \theta_1) = \tau_z \cdot \Delta\theta$

Work and Power in Rotational Motion

- When a torque does work on a rotating rigid body, the kinetic energy changes by an amount equal to the work done.
- We can prove this by using exactly the same procedure that we used in Lecture 7 – Work-Energy Theorem.

$\frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$

扭矩做功

Total work done on a rotating rigid body = work done by the net external torque

$$W_{\text{tot}} = \int_{\omega_1}^{\omega_2} I\omega_z\, d\omega_z = \underbrace{\tfrac{1}{2}I\omega_2^2}_{\text{Final rotational kinetic energy}} - \underbrace{\tfrac{1}{2}I\omega_1^2}_{\text{Initial rotational kinetic energy}} \quad (10.22)$$

- This equation is analogous to the work–energy theorem for a particle.

相似的.

Work and Power in Rotational Motion

- The change in the rotational kinetic energy of a rigid body equals the work done by forces exerted from outside the body.
- The rotational kinetic energy of a helicopter's main rotor is equal to the total work done to set it spinning.
- When it is spinning at a constant rate, positive work is done on the rotor by the engine and negative work is done on it by air resistance.
- Hence the net work being done is **zero** and the kinetic energy remains **constant**.



Work and Power in Rotational Motion

- To find the relation of power and torque, we divide both sides of equation $dW = \tau_z \, d\theta$ by the time interval *dt* during which the angular displacement occurs, we find

$$\left(\frac{dW}{dt}\right) = \left(\tau_z \frac{d\theta}{dt}\right) \quad W_z$$

- Since *dW/dt* is the rate of doing work, or **power P**, and *dθ/dt* is the angular velocity ω_z:

Power due to a torque acting on a rigid body

$$P = \tau_z \omega_z \qquad (10.23)$$

Torque with respect to body's rotation axis

Angular velocity of body about axis

$W = FS$

$\rightarrow P = Fv$

Work and Power in Rotational Motion

EXAMPLE 10.8 CALCULATING POWER FROM TORQUE

An electric motor exerts a constant 10-N · m torque on a grindstone, which has a moment of inertia of 2.0 kg · m^2 about its shaft. The system starts from rest. Find the work W done by the motor in 8.0 s and the grindstone's kinetic energy K at this time. What average power P_{av} is delivered by the motor?

SOLUTION

IDENTIFY and SET UP: The only torque acting is that due to the motor. Since this torque is constant, the grindstone's angular acceleration α_z is constant. We'll use Eq. (10.7) to find α_z, and then use this in the kinematics equations from Section 9.2 to calculate the angle $\Delta\theta$ through which the grindstone rotates in 8.0 s and its final angular velocity ω_z. From these we'll calculate W, K, and P_{av}.

EXECUTE: We have $\sum\tau_z = 10\ \text{N}\cdot\text{m}$ and $I = 2.0\ \text{kg}\cdot\text{m}^2$, so $\sum\tau_z = I\alpha_z$ yields $\alpha_z = 5.0\ \text{rad/s}^2$. From Eqs. (9.11) and (10.21),

$$\Delta\theta = \tfrac{1}{2}\alpha_z t^2 = \tfrac{1}{2}(5.0\ \text{rad/s}^2)(8.0\ \text{s})^2 = 160\ \text{rad}$$

$$W = \tau_z \Delta\theta = (10\ \text{N}\cdot\text{m})(160\ \text{rad}) = 1600\ \text{J}$$

From Eqs. (9.7) and (9.17),

$$\omega_z = \alpha_z t = (5.0\ \text{rad/s}^2)(8.0\ \text{s}) = 40\ \text{rad/s}$$

$$K = \tfrac{1}{2} I\omega_z^2 = \tfrac{1}{2}(2.0\ \text{kg}\cdot\text{m}^2)(40\ \text{rad/s})^2 = 1600\ \text{J}$$

The average power is the work done divided by the time interval:

$$P_{av} = \frac{1600\ \text{J}}{8.0\ \text{s}} = 200\ \text{J/s} = 200\ \text{W}$$

EVALUATE: The initial kinetic energy was zero, so the work done W must equal the final kinetic energy K [Eq. (10.22)]. This is just as we calculated. We can check our result $P_{av} = 200$ W by considering the *instantaneous* power $P = \tau_z\omega_z$. Because ω_z increases continuously, P increases continuously as well; its value increases from zero at $t = 0$ to $(10\ \text{N}\cdot\text{m})(40\ \text{rad/s}) = 400$ W at $t = 8.0$ s. Both ω_z and P increase *uniformly* with time, so the *average* power is just half this maximum value, or 200 W.

$\frac{1}{2} \times 2 \times 40^2 = 1600$ $w = 40\ rad/s$

Angular Momentum 角动量

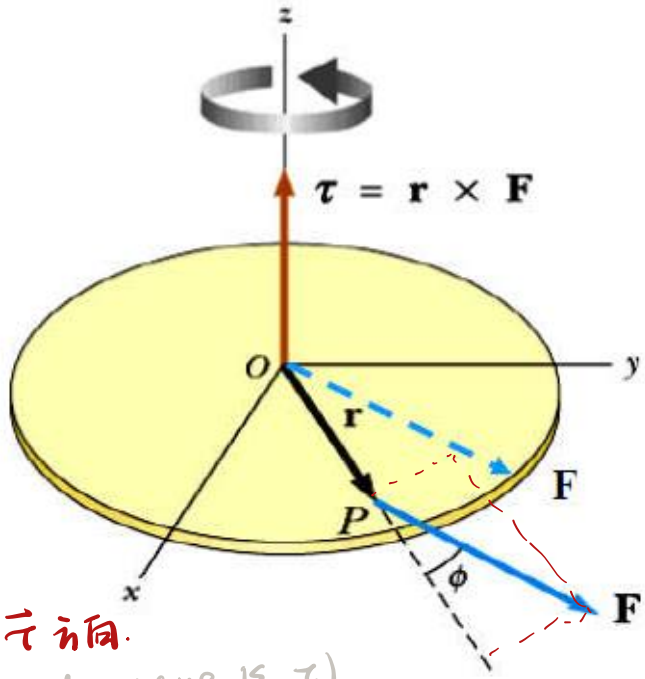
- Torque
  - Remember that the torque of a force was earlier defined as:

$$\tau = Fl = rF\sin\theta$$

  - This is just the magnitude of torque vector about O:

$$\vec{\tau} = \vec{r} \times \vec{F}$$

  - Direction of vector is normal to the planed formed by **r** and **F**.



右手定则得方向.

(单纯 magnitude of torque is. τ)

角动量守恒定律是物理学的普遍定律之一，反映质点和质点系围绕一点或一轴运动的普遍规律；反映不受外力作用或所受诸外力对某定点（或定轴）的合力矩始终等于零的质点和质点系围绕该点（或轴）运动的普遍规律。

角动量守恒定律是对于质点，角动量定理可表述为质点对固定点的角动量对时间的微商，等于作用于该质点上的力对该点的力矩。

Angular Momentum 引入角动量 → L

- Momentum of a particle is given by

Momentum of a particle (a vector quantity) $\vec{p} = m\vec{v}$ (Particle mass, Particle velocity)

Newton's second law in terms of momentum: The net force acting on a particle ... $\Sigma\vec{F} = \frac{d\vec{p}}{dt}$ ... equals the rate of change of the particle's momentum.

F = 动量变化量

类似 动量

- The analog of momentum of a particle is angular momentum, a vector quantity denoted as **L**:

cross product

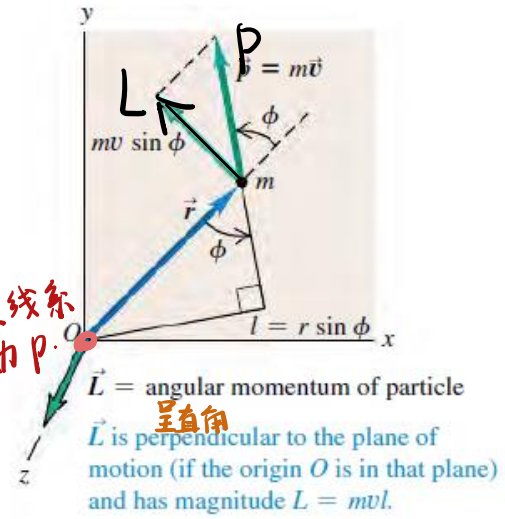
Angular momentum of a particle relative to origin O of an inertial frame of reference

$$\vec{L} = \vec{r} \times \vec{p} = \vec{r} \times m\vec{v} \qquad (10.24)$$

($\vec{r}$: Position vector of particle relative to O; $\vec{p}$: Linear momentum of particle = mass times velocity)

- The angular momentum vector **L** is perpendicular to the xy-plane.

$$L = mvr\sin\phi = mvl \qquad (10.25)$$



不完全，这里expression是线系 → 统表达圆周运动 p.

垂直角

- Momentum is not covered in this course but you can read it on chapter 8 of the textbook.

P=mv L=mvl 因为在角体系中总是与点到圆心距离有关系的

Angular Momentum

- Relationship of torque and angular momentum

$$\frac{d\vec{L}}{dt} = \frac{d}{dt}(\vec{r} \times \vec{p}) = \vec{r} \times \frac{d\vec{p}}{dt} + \frac{d\vec{r}}{dt} \times \vec{p}$$

But $\frac{d\vec{r}}{dt} \times \vec{p} = \vec{v} \times m\vec{v} = 0$

$$\frac{d\vec{L}}{dt} = \vec{r} \times \frac{d\vec{p}}{dt}$$

We know that $\sum \vec{F} = \frac{d\vec{p}}{dt}$

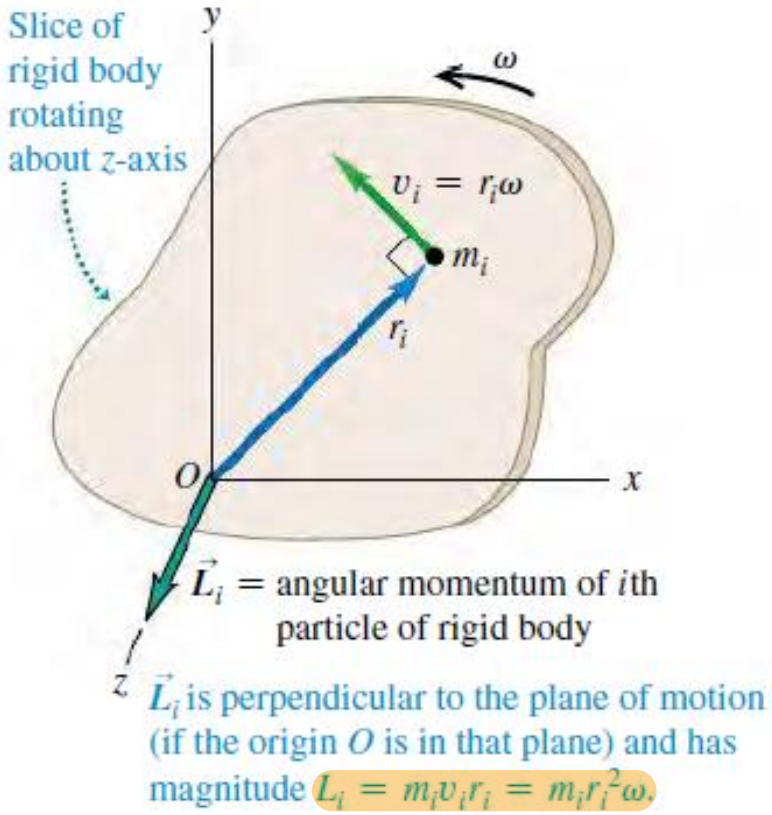
$$\vec{\tau} = \vec{r} \times \vec{F} = \vec{r} \times \frac{d\vec{p}}{dt}$$

$$\vec{\tau} = \frac{d\vec{L}}{dt} \quad (10.26)$$

The rate of change of angular momentum of a particle equals the torque of the net force acting on it.

Angular Momentum

- Consider rotation of a rigid object about a fixed axis.



For a particle with mass m_i,

$$L_i = m_i r_i^2 \omega \quad (10.27)$$

For the whole object

$$L_z = \sum_i m_i r_i^2 \omega$$

$$L_z = I\omega \quad (10.28)$$

Angular Momentum

- Differentiate $L_z = I\omega$ with respect to time

$$\frac{dL_z}{dt} = I\frac{d\omega}{dt} = I\alpha$$

$$\sum \tau_{\text{ext}} = \frac{dL_z}{dt} = I\alpha \qquad (10.29)$$

- The net external torque acting on a rigid object rotating about a fixed axis equals the product of the moment of inertia and the angular acceleration about that axis.

Angular Momentum

Example 10.9 **Angular momentum and torque**

A turbine fan in a jet engine has a moment of inertia of 2.5 kg · m² about its axis of rotation. As the turbine starts up, its angular velocity is given by $\omega_z = (40\ \text{rad/s}^3)t^2$. (a) Find the fan's angular momentum as a function of time, and find its value at $t = 3.0$ s. (b) Find the net torque on the fan as a function of time, and find its value at $t = 3.0$ s.

Solution:

a) $$L_z = I\omega_z = (2.5\ \text{kg} \cdot \text{m}^2)(40\ \text{rad/s}^3)t^2 = (100\ \text{kg} \cdot \text{m}^2/\text{s}^3)t^2$$

At $t = 3.0$ s, $L_z = 900\ \text{kg} \cdot \text{m}^2/\text{s}$

$$\tau_z = \frac{dL_z}{dt} = (100\ \text{kg} \cdot \text{m}^2/\text{s}^3)(2t) = (200\ \text{kg} \cdot \text{m}^2/\text{s}^3)t$$

At $t = 3.0$ s,

$$\tau_z = (200\ \text{kg} \cdot \text{m}^2/\text{s}^3)(3.0\ \text{s}) = 600\ \text{kg} \cdot \text{m}^2/\text{s}^2 = 600\ \text{N} \cdot \text{m}$$

Conservation of Angular Momentum

- The total angular momentum of a system is constant if the resultant external torque acting on the system is zero.

$$\sum \vec{\tau}_{\text{ext}} = \frac{d\vec{L}}{dt} = 0 \quad \Rightarrow \quad \vec{L} = \text{const.}$$

- For a system of particles: $\sum \vec{L}_i = \text{const.}$
- Conservation of angular momentum: $\vec{L}_i = \vec{L}_f$
- For rotation about a fixed axis: $I_i\omega_i = I_f\omega_f$ (10.30)
- The above is valid for rotation about a fixed axis or axis through the centre of mass of a moving system as long as the axis remains fixed in direction.

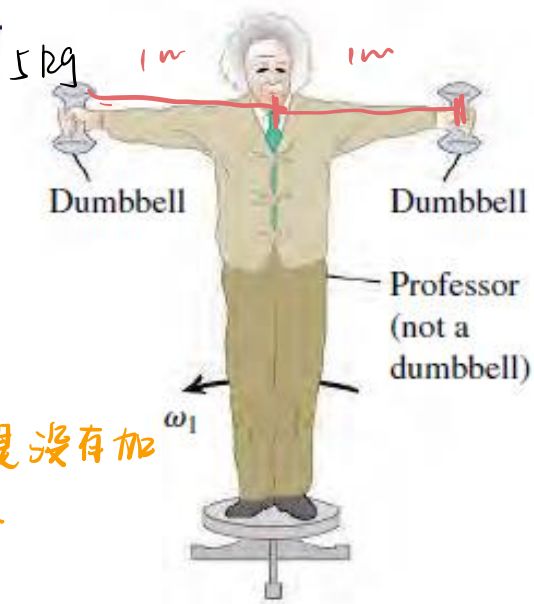
Conservation of Angular Momentum

EXAMPLE 10.10 ANYONE CAN BE A BALLERINA

A physics professor stands at the center of a frictionless turntable with arms outstretched and a 5.0-kg dumbbell in each hand (**Fig. 10.29**). He is set rotating about the vertical axis, making one revolution in 2.0 s. Find his final angular velocity if he pulls the dumbbells inward to his stomach. His moment of inertia (without the dumbbells) is 3.0 kg · m^2 with arms outstretched and 2.2 kg · m^2 with his hands at his stomach. The dumbbells are 1.0 m from the axis initially and 0.20 m at the end.

$\theta = 2\pi$ $\quad w = \frac{\theta}{t} = \frac{2\pi}{2} = \pi$

要注意的是，题中I是没有加上两个 dumbbells 的.



BEFORE



AFTER

$I_1 w_1 = I_2 w_2$

$3 \times \pi = 2.2 \cdot w_2$

$\therefore w_2 = 4.28$ rad/s

Conservation of Angular Momentum

SOLUTION

IDENTIFY, SET UP, and EXECUTE: No external torques act about the z-axis, so L_z is constant. We'll use Eq. (10.30) to find the final angular velocity ω_{2z}. The moment of inertia of the system is $I = I_{\text{prof}} + I_{\text{dumbbells}}$. We treat each dumbbell as a particle of mass m that contributes mr^2 to $I_{\text{dumbbells}}$, where r is the perpendicular distance from the axis to the dumbbell. Initially we have

$$I_1 = 3.0\ \text{kg}\cdot\text{m}^2 + 2(5.0\ \text{kg})(1.0\ \text{m})^2 = 13\ \text{kg}\cdot\text{m}^2$$

$$\omega_{1z} = \frac{1\ \text{rev}}{2.0\ \text{s}} = 0.50\ \text{rev/s}$$

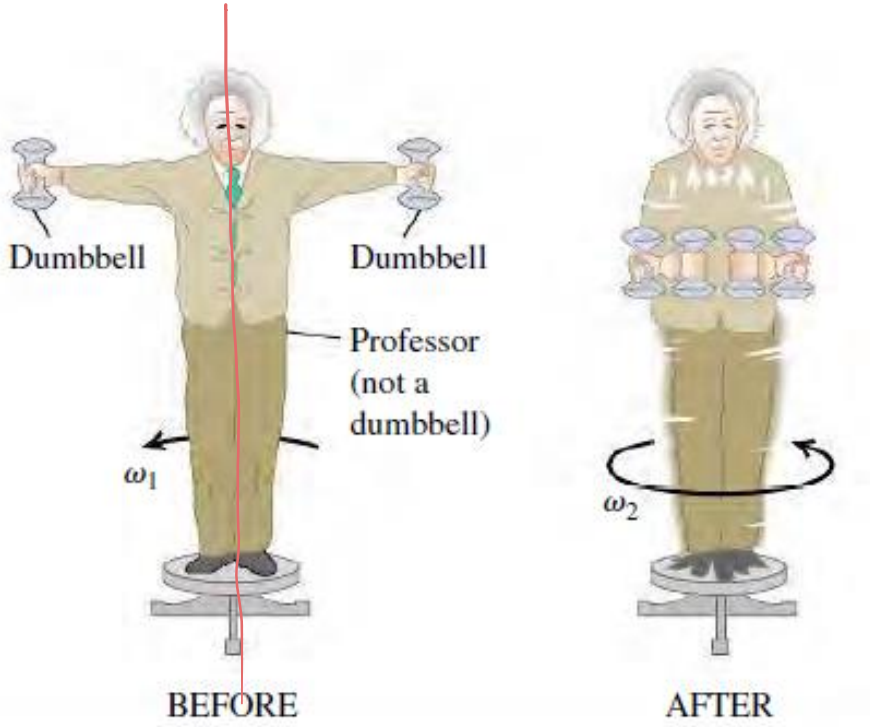
The final moment of inertia is

$$I_2 = 2.2\ \text{kg}\cdot\text{m}^2 + 2(5.0\ \text{kg})(0.20\ \text{m})^2 = 2.6\ \text{kg}\cdot\text{m}^2$$

From Eq. (10.30), the final angular velocity is

$$\omega_{2z} = \frac{I_1}{I_2}\omega_{1z} = \frac{13\ \text{kg}\cdot\text{m}^2}{2.6\ \text{kg}\cdot\text{m}^2}(0.50\ \text{rev/s}) = 2.5\ \text{rev/s} = 5\omega_{1z}$$

Can you see why we didn't have to change "revolutions" to "radians" in this calculation?



No external torques

$$I_1\omega_1 = I_2\omega_2$$

Conservation of Angular Momentum

EVALUATE: The angular momentum remained constant, but the angular velocity increased by a factor of 5, from $\omega_{1z} = (0.50 \text{ rev/s}) \times (2\pi \text{ rad/rev}) = 3.14 \text{ rad/s}$ to $\omega_{2z} = (2.5 \text{ rev/s})(2\pi \text{ rad/rev}) = 15.7 \text{ rad/s}$. The initial and final kinetic energies are then

$$K_1 = \tfrac{1}{2} I_1 {\omega_{1z}}^2 = \tfrac{1}{2}(13 \text{ kg} \cdot \text{m}^2)(3.14 \text{ rad/s})^2 = 64 \text{ J}$$

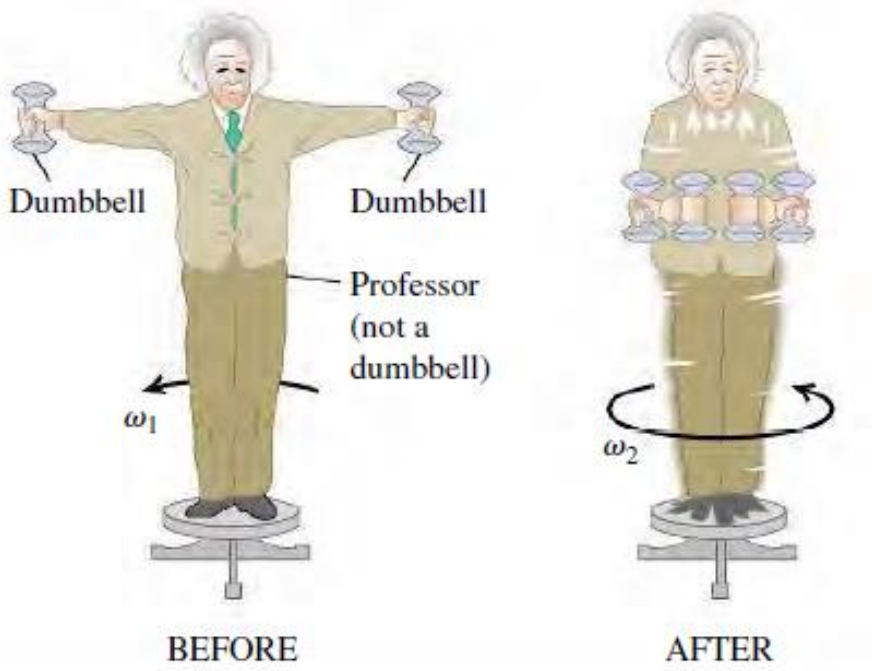
$$K_2 = \tfrac{1}{2} I_2 {\omega_{2z}}^2 = \tfrac{1}{2}(2.6 \text{ kg} \cdot \text{m}^2)(15.7 \text{ rad/s})^2 = 320 \text{ J}$$

$$K_2 - K_1 > 0$$

Professor has to do +ve work ! Why?

In his frame, there are forces pulling the dumbbells outwards (we called centrifugal force in our earlier lectures), and he has to apply an inward force to pull the dumbbells in. Force he exerted on dumbbells is in the direction of displacement of dumbbells ⇒ +ve work.

The fivefold increase in kinetic energy came from the work that the professor did in pulling his arms and the dumbbells inward.



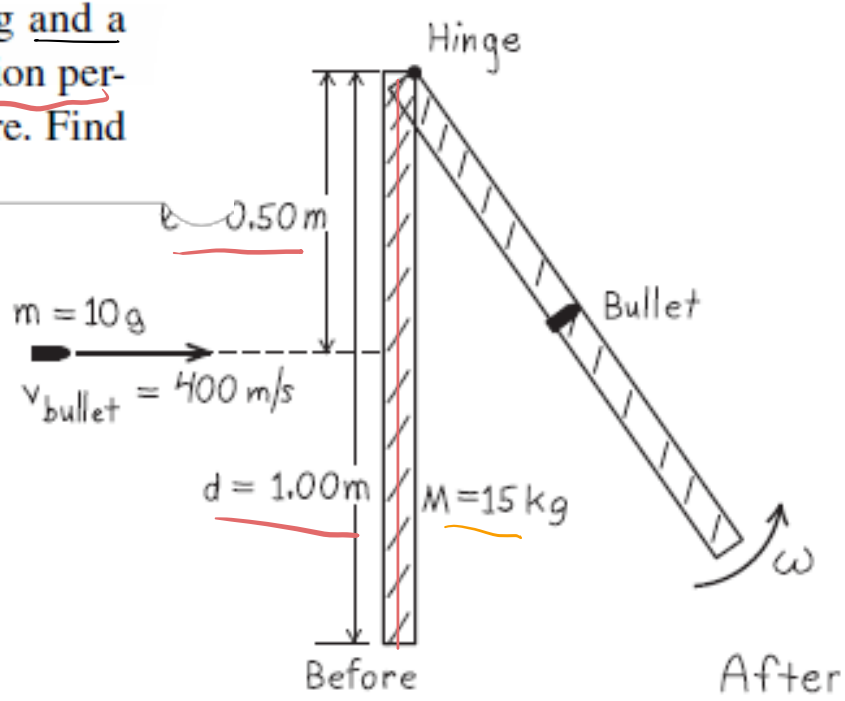
No external torques

$$I_1\omega_1 = I_2\omega_2$$

Conservation of Angular Momentum

EXAMPLE 10.12 ANGULAR MOMENTUM IN A CRIME BUST

A door 1.00 m wide, of mass 15 kg, can rotate freely about a vertical axis through its hinges. A bullet with a mass of 10 g and a speed of 400 m/s strikes the center of the door, in a direction perpendicular to the plane of the door, and embeds itself there. Find the door's angular speed. Is kinetic energy conserved?



Conservation of Angular Momentum

Solution:

bullet

$$L = mvl = (0.010\ \text{kg})(400\ \text{m/s})(0.50\ \text{m}) = 2.0\ \text{kg}\cdot\text{m}^2/\text{s}$$

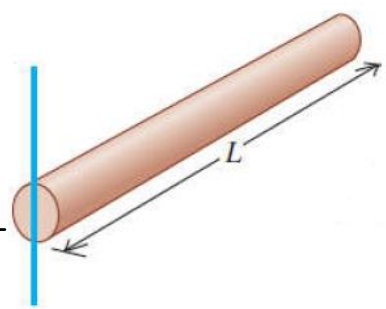
The final angular momentum is $I\omega$.

$I = I_{\text{door}} + I_{\text{bullet}}$.

$I = mr^2$

(b) Slender rod, axis through one end

$$I = \tfrac{1}{3}ML^2$$



$$I_{\text{door}} = \frac{Md^2}{3} = \frac{(15\ \text{kg})(1.00\ \text{m})^2}{3} = 5.0\ \text{kg}\cdot\text{m}^2$$

$$I_{\text{bullet}} = ml^2 = (0.010\ \text{kg})(0.50\ \text{m})^2 = 0.0025\ \text{kg}\cdot\text{m}^2$$

Conservation of angular momentum requires that $mvl = I\omega$,

$$\omega = \frac{mvl}{I} = \frac{2.0\ \text{kg}\cdot\text{m}^2/\text{s}}{5.0\ \text{kg}\cdot\text{m}^2 + 0.0025\ \text{kg}\cdot\text{m}^2} = 0.40\ \text{rad/s}$$

The initial and final kinetic energies are

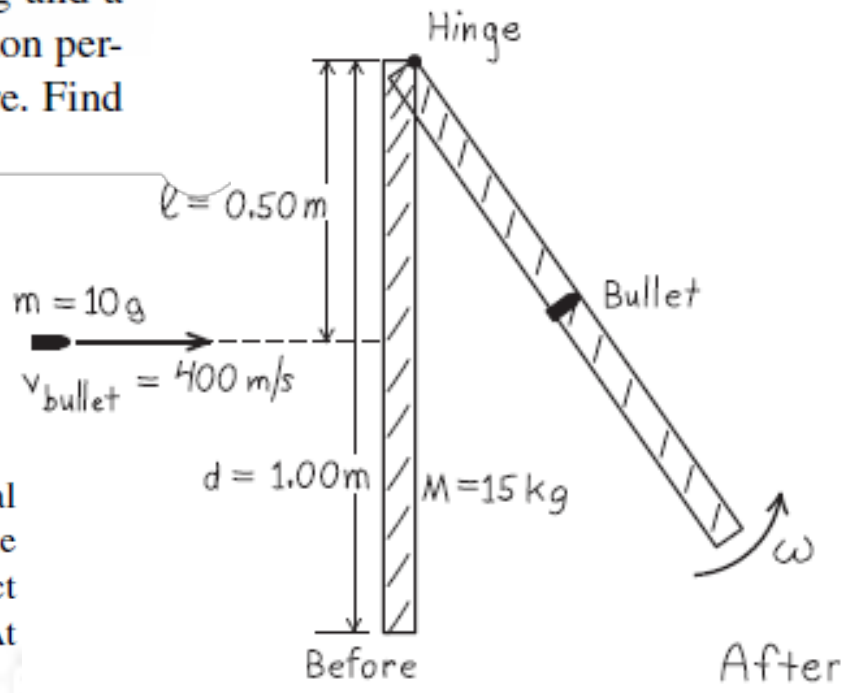
$$K_1 = \tfrac{1}{2}mv^2 = \tfrac{1}{2}(0.010\ \text{kg})(400\ \text{m/s})^2 = 800\ \text{J}$$

$$K_2 = \tfrac{1}{2}I\omega^2 = \tfrac{1}{2}(5.0025\ \text{kg}\cdot\text{m}^2)(0.40\ \text{rad/s})^2 = 0.40\ \text{J}$$

Conservation of Angular Momentum

EXAMPLE 10.12 ANGULAR MOMENTUM IN A CRIME BUST

A door 1.00 m wide, of mass 15 kg, can rotate freely about a vertical axis through its hinges. A bullet with a mass of 10 g and a speed of 400 m/s strikes the center of the door, in a direction perpendicular to the plane of the door, and embeds itself there. Find the door's angular speed. Is kinetic energy conserved?



The initial and final kinetic energies are

$$K_1 = \tfrac{1}{2}mv^2 = \tfrac{1}{2}(0.010\text{ kg})(400\text{ m/s})^2 = 800\text{ J}$$

$$K_2 = \tfrac{1}{2}I\omega^2 = \tfrac{1}{2}(5.0025\text{ kg}\cdot\text{m}^2)(0.40\text{ rad/s})^2 = 0.40\text{ J}$$

EVALUATE: The final kinetic energy is only $\frac{1}{2000}$ of the initial value! We did not expect kinetic energy to be conserved: The collision is inelastic because nonconservative friction forces act during the impact. The door's final angular speed is quite slow: At 0.40 rad/s, it takes 3.9 s to swing through 90° ($\pi/2$ radians).

Conservation of Angular Momentum

- Conservation of energy

$$K_i + U_i = K_f + U_f$$

- Conservation of linear momentum

$$\vec{p}_i = \vec{p}_f$$

- Conservation of angular momentum

$$\vec{L}_i = \vec{L}_f$$

For any isolated system

Summary

- Dynamics of Rotational Motion has been studied.
  - Basic concepts and principles are discussed with detailed analysis and examples
    1. **Torque**
    2. **Torque and Angular Acceleration**
    3. **Rigid-Body Rotation About a Moving Axis**
    3. **Work and Power in Rotational Motion**
    4. **Angular Momentum**
- Textbook
  - Chapter 10：Dynamics of Rotational Motion in the textbook.